Knowledge Graphs and Fine-Grained Visual Features: A Potent Duo Against Cheapfakes

Tuan-Vinh La
MMLab
University of Information Technology
Ho Chi Minh, Vietnam
vinhlt.16@grad.uit.edu.vn

Minh-Hieu Nguyen
University of Science
Ho Chi Minh, Vietnam
nmhieu@apcs.fitus.edu.vn

Minh-son Dao
National Institute of Information and Communications Technology
Tokyo, Japan
dao@nict.go.jp

Abstract

The digital era has revolutionized information sharing but has also fueled the rapid spread of misinformation, particularly through cheapfakes—low-effort manipulations combining text and visuals to mislead audiences. Their ease of creation makes them a prevalent tool for deception on social media, significantly shaping public perception. Existing detection methods rely on pre-trained datasets but suffer from limited diversity, weak contextual understanding, and inadequate multimodal reasoning. To address these challenges, we propose a novel framework that: (1) integrates external knowledge through knowledge graphs to enhance contextual comprehension, (2) extracts fine-grained visual features to better capture subtle manipulations, and (3) leverages multimodal learning for improved text-image consistency analysis. Unlike prior approaches, our method generalizes across multiple datasets, demonstrating robust adaptability and higher accuracy in misinformation detection. Extensive experiments on real-world datasets validate its effectiveness, offering a scalable and reliable solution for combating cheapfakes.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; **Computer vision**.

Keywords

Deep Learning, Computer Vision, Natural Language Processing, Knowledge Graph, Cheapfakes, Misinformation, News

ACM Reference Format:
Tuan-Vinh La, Minh-Hieu Nguyen, and Minh-son Dao. 2025. Knowledge Graphs and Fine-Grained Visual Features: A Potent Duo Against Cheapfakes. In *Proceedings of the 27th International Conference on Multimodal Interaction (ICMI '25), October 13–17, 2025, Canberra, ACT, Australia.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3716553.3750812

1 Introduction

The rise of the Internet and social media has revolutionized information exchange but also fueled the spread of fake news, with cheapfakes emerging as a prevalent form. These low-cost manipulations involve simple alterations or misaligned images and text, creating deceptive narratives across politics, science, and celebrity culture. Their ease of creation has led to widespread dissemination, significantly shaping public perception [13, 29].

Verifying cheapfakes is challenging as it requires analyzing both textual and visual elements in context. Traditional fact-checking often demands domain expertise and specialized tools, making manual detection difficult. Studies show that human accuracy in identifying out-of-context media without search assistance is only 65% [24].

Generally, fake news encompasses a variety of types, each with distinct attributes and categories. Moreover, in numerous benchmark datasets, fake news often exhibits multiple types, requiring a general solution to address them comprehensively. In this paper, we focus specifically on addressing a particular type of fake news - cheapfakes. Despite their widespread presence and impact on both social media platforms and reality, research on and benchmark datasets for cheapfakes detection are currently limited.

To the best of our knowledge, COSMOS [3] represents the first initiative to provide both a dataset and a solution for the domain of cheapfakes detection. The COSMOS benchmark dataset was sourced from two primary channels: news websites[1] and fact-checking websites. This benchmark was introduced as part of the ACMMM'22 grand challenge and features two distinct tasks. Both tasks require participants to assess the truthfulness of a given caption and its associated image. The difference lies in the evaluation set. In Task 1, each pair comprises two captions and one attached image (image, caption$_1$, caption$_2$), while Task 2 consists of only one caption and one attached image (image, caption). A notable challenge is that the training dataset does not contain false context (out-of-context), making the challenge more difficult to address.

Previous research on cheapfakes detection has centered around exploiting multi-modal methods, integrating various information representations extracted from large-scale pre-trained datasets, and leveraging characteristics of the evaluation dataset. Despite their promising results, these methods exhibit three key limitations:

(1) **Ambiguous Image-Caption Semantics:** News media often feature weakly aligned image-caption pairs. For instance, Image 2 shows people at a table, while the caption refers to "election workers," creating subtle context gaps. Traditional encoders struggle to model such fine-grained relationships due to poor recognition of small or diverse objects.

[1]New York Times, CNN, Reuters, ABC, PBS, NBCLA, AP News, Sky News, Telegraph, Time, DenverPost, Washington Post, CBC News, Guardian, Herald Sun, Independent, CS Gazette, BBC

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
ICMI '25, Canberra, ACT, Australia
© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-1499-3/25/10
https://doi.org/10.1145/3716553.3750812

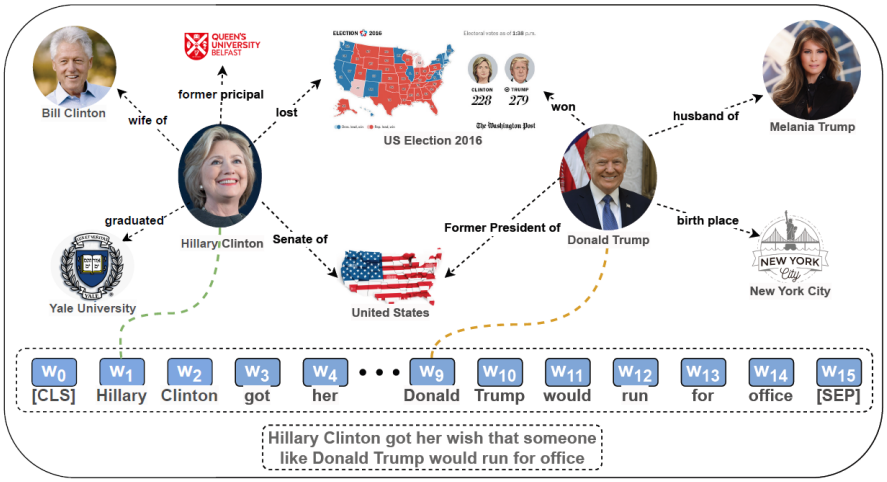
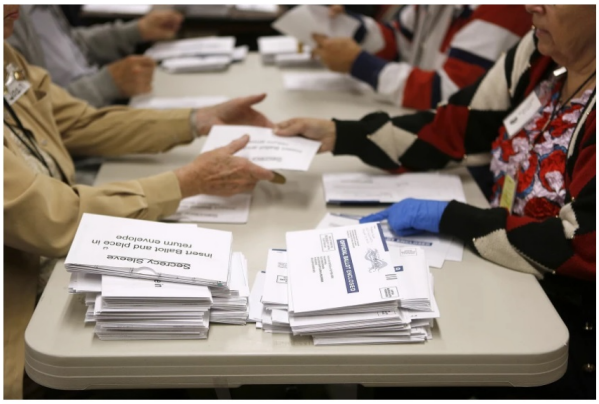


Figure 1: The knowledge graph embeddings capture complex entity relationships, enhancing the model's ability to understand nuanced connections within the information. This integration of external knowledge improves contextual awareness and enriches textual comprehension.

(2) **Lack of External Knowledge:** Standard language models (e.g., BERT-uncased with 30k tokens) often fail to recognize uncommon or ambiguous entities. For example, "Don Moen" may be treated as unknown, and "Donald" could map to "Donald Duck" instead of "Donald Trump," leading to semantic misinterpretation.

(3) **Overfitting to Dataset-Specific Rules:** Prior work often relies on handcrafted or dataset-tuned strategies, reducing model generalizability across different misinformation scenarios.



Election workers in Colorado open mail-in ballots. (Marc Piscotty / Getty Images)

Figure 2: Example of ambigous relation between caption and image. Traditional image encoders tend to ignore such fine-grained relationships due to their limited recognition of small objects or lack of diverse category representations in training datasets

To address the limitations mentioned above,approach provides innovative solutions. It incorporates interpretable reasoning for visual feature extraction, seamlessly integrates external knowledge into textual feature extraction, and extends experiments to additional benchmarks to reinforce the effectiveness of our proposal.

We introduce a groundbreaking approach aimed at overcoming the three key limitations identified in previous research. To address limitations (1) and (2), we present two distinct frameworks for extracting visual and textual features, respectively.

For visual feature extraction, inspired by Anderson et al.[2], we propose an interpretable reasoning model that effectively extracts and interacts with fine-grained region objects in images. This novel approach enables us to surpass the limitations of limited recognition of high-level semantic concepts and the focus on local, pixel-level analysis in standard methods.

To address the second limitation, we introduce a framework that ingeniously integrates external knowledge into our language representation model. To mitigate issues arising from standard vocabularies, we utilize a knowledge graph [6] containing over 5 million entities and 20 million triplets, dedicated to memorizing and reasoning about diverse entity relationships. This rich source of information is seamlessly incorporated into our language representation model, significantly enhancing its ability to comprehend context and deepen its understanding of textual content.

As illustrated in Figure 1, the entity embeddings derived from this knowledge graph adeptly capture intricate relations between entities, facilitating a more comprehensive understanding of abstract connections between multiple entities in the provided information. This integration of external knowledge greatly enriches our model's contextual comprehension and elevates its understanding of textual content.

In addition to addressing restrictions (1) and (2), we also tackled the limitations of conventional research in restriction (3). Rather than evaluating our approach solely on the COSMOS dataset, we

expanded our methodology to include other fake news datasets that incorporate both visual and textual information for a comprehensive assessment. To this end, we utilized the FakenewsNet dataset [33], a well-established benchmark in fake news detection. Notably, this dataset includes not only visual and textual information but also social context data, enriching the depth of our evaluation.

The significant contributions of our work can be summarized as follows:

(1) **Fine-Grained Visual and Knowledge Integration:** We are the first to explore fine-grained object regions for cheapfake detection and introduce a novel framework that integrates knowledge graphs into language modeling. This reduces reliance on shallow linguistic features and enhances semantic understanding.
(2) **Extensive Evaluation Across Datasets:** Our method is validated on multiple real-world datasets through rigorous experimentation, confirming its robustness and generalizability.
(3) **State-of-the-Art Results:** The approach achieves state-of-the-art accuracy on COSMOS (Task 1 and Task 2) and performs competitively on FakenewsNet, demonstrating its effectiveness.

Our pioneering approach marks significant strides in the realm of cheapfakes detection. By harnessing the power of a knowledge graph and adopting a versatile evaluation approach, we not only overcome the limitations of previous research but also achieve competitive performance levels within the field. This contribution strengthens the foundations of cheapfakes detection and enhances its real-world applicability.

This work prioritizes ethical considerations, focusing on data privacy, bias mitigation, and inclusivity. We address potential risks and misuse, ensuring responsible deployment of our technology. Our approach reflects a commitment to safe and responsible innovation, considering the broader societal impact of multimodal interactions.

2 Related Works

In this section, we present a brief overview of Cheapfakes Detection and related domains, highlighting their strengths and weaknesses. This overview serves to reinforce our motivation to introduce our novel method.

2.1 Fake News Detection

The realm of multimedia has witnessed extensive exploration in the detection of fake news, historically focusing on textual formats. Early works, such as those by Shu et al. [33] and Conroy et al. [10] focused on hand-crafted patterns and mining perspectives to verify the veracity of information.

However, the advent of deep learning has opened a new era. Research like that of Gundapu [14] has utilized novel frameworks such as Transformer [39] to learn lexical, syntactic, semantic, and linguistic features.

Wang et. al. [43] have analyzed the content, role, and characteristics of visual and textual information. This analysis is grounded in the observation that news articles and posts often include both textual information and one or more attached images. These studies attempted to integrate textual and visual features to learn the multimodal representation of news content. For instance, Spotfake [36] extracted textual and visual representations from a large pre-trained dataset, then concatenated them to classify news. EANN [43] introduced an additional event discriminator through an adversarial network, while MVAE [17] incorporated a reconstruction layer to enhance the discovery of correlations across different modalities. To elevate the understanding of content, approaches like [21] harnessed image captioning to describe visual information. Research [1] utilizes online search tools to retrieve relevant news and extract web context, ensuring consistency between original and retrieved information. Additionally, studies such as [28] investigate the performance of large language models (LLMs) in fake news detection.

Despite achieve notable results, existing multimodal fake news detection methods face notable limitations. Multimodal-based approaches often struggle with fine-grained semantic alignment between text and images, making them ineffective in detecting subtle contextual inconsistencies such as mismatched objects or misleading captions. These models primarily rely on statistical correlations rather than factual reasoning, limiting their ability to verify information against real-world knowledge. Similarly, search-based verification tools, while useful for retrieving external evidence, are constrained by indexed content availability, making them ineffective for dealing with events where reliable sources are not available. Moreover, they primarily focus on text retrieval, lacking the ability to analyze multimodal inconsistencies between images and caption [4]. Large language models (LLMs) further expose weaknesses in misinformation detection due to their static knowledge base, which prevents them from adapting to evolving news cycles. Additionally, they are prone to hallucinations, often generating convincing yet factually incorrect information due to their reliance on probabilistic language patterns rather than structured factual reasoning [7]. LLMs also lack interpretability, making it difficult to trace how a claim is verified, and they remain susceptible to adversarial manipulation through engineered prompts. In contrast, knowledge graphs (KGs) can improve these limitations by offering structured, updatable knowledge that enables fact-driven verification rather than pattern-based classification. KGs establish explicit relationships between entities, allowing for cross-modal consistency checks that can detect subtle discrepancies between text and images. By integrating knowledge graphs into fake news detection, we bridge the gap between pattern recognition in deep learning, evidence retrieval in search tools, and contextual analysis in LLMs, creating a more robust, fact-driven approach to combating misinformation.

2.2 Cheapfakes Detection

Cheapfakes detection has been explored through various methodologies, with early research primarily relying on pretrained language models and self-supervised learning. The COSMOS benchmark [3] introduced a dataset and baseline approach leveraging a pretrained language model to assess conflicting information between two captions. This method also utilized self-supervised learning to align captions with objects in images, aiming to improve contextual understanding. Subsequent research [38] enhanced detection performance by analyzing handcrafted patterns from evaluation datasets. Others [20, 21] employed dataset-specific boosting techniques,

assessing performance on subsets of the dataset. While these methods demonstrated improvement, their reliance on dataset-specific characteristics limits their generalizability to real-world scenarios. Recent approaches addressed these limitations by enhancing contextual understanding and multimodal alignment. Vo-Hoang et al. [40] proposed a method combining LLaMa-Adapter V2 and RoBERTa for detecting out-of-context media, demonstrating improved accuracy through transformer-based contextual learning. Pham et al. [27] introduced a Generative Adaptive Context Learning Framework, allowing large language models to dynamically adjust to contextual variations, thereby improving cheapfake detection. Le et al. developed TeGA [23], a text-guided generative-based approach that refines multimodal representation learning using textual guidance. Vu et al. [41] advanced cheapfake detection by incorporating prompt engineering with interleaved text-image models, improving the interpretability and performance of multimodal classifiers. Additionally, Seo et al. [31] proposed a multi-stage deep learning approach integrating text-image and image-image comparisons to refine semantic alignment in cheapfakes detection.

Despite the contributions of these studies, key challenges remain. Many existing methods rely on dataset-dependent features, limiting their applicability to unseen misinformation scenarios. Moreover, traditional multimodal approaches struggle with fine-grained object recognition and semantic reasoning [22], making them susceptible to misleading out-of-context manipulations.

Our proposed method addresses these shortcomings by integrating knowledge graphs with fine-grained visual feature extraction, enabling a more comprehensive understanding of the relationships between textual and visual elements. Unlike previous works that depend on static dataset characteristics, our approach ensures generalizability by not incorporating evaluation dataset features during training. Furthermore, we expand our evaluation beyond COSMOS to include additional fake news datasets, assessing the robustness and adaptability of our method across different misinformation contexts. By enhancing both semantic reasoning and multimodal alignment, our approach significantly improves the reliability of cheapfake detection compared to prior methodologies.

3 Methodology

In this section, we offer an extensive overview of our approach to cheapfakes detection. Our primary objective is to explore an effective methodology for extracting and integrating the relationships between sentence semantics, knowledge graphs, and fine-grained visual features. To achieve this, we introduce a comprehensive framework that places a distinct focus on three critical aspects:

(1) Integration of Knowledge Graph with Sentence Semantic Embedding.
(2) Extraction of Fine-Grained Visual Features.
(3) Integration of Visual and Textual Results for Predicting the Final Output.

Problem Definition and Key Notation: In our context, we are provided with a news article denoted as A = {T,I}, where T represents the textual information, and I signifies the attached image, which conveys visual information. We also introduce a knowledge graph, denoted as $G = \{\mathcal{E}, \mathcal{R}\}$, which comprises the sets of entity embeddings, represented as $\mathcal{E}$, and relation embeddings, represented as $\mathcal{R}$.

3.1 Textual Features Extraction

Given textual information T, we first use subword tokenizer to tokenize T to sequence of embedding token $W = \{w_0, ..., w_n\}, w \in V, R^{D_w}$, where n is length of token, V is token vocabulary, and R^{D_w} is dimension of each word embedding token.

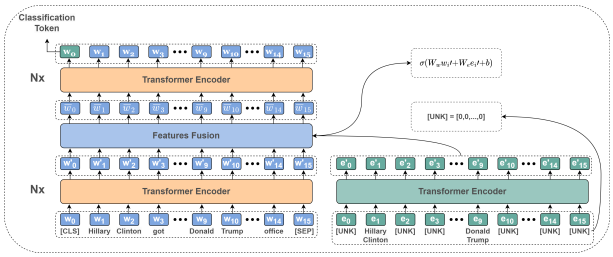


Figure 3: An Overview of the Textual Features Extraction Framework. Initially, both the textual token sequence and entity token sequence undergo inference using a Transformer Encoder. Subsequently, a feature fusion layer integrates knowledge into the textual context, followed by a feed-forward process through an additional Transformer Encoder to attain the final representation. The initial token [CLS] is designated as the classification token.

We also employ tagme[2] to extract entities present in the textual information and link them to their respective Wikipedia entries. We represent the entity sequence tokens as $E = e_0, ..., e_n, e \in \mathcal{E}, R^{D_e}$, where R^{D_e} signifies the dimension of each entity embedding token. Each entity, denoted as e, is aligned with the first token of the embedding word. In cases where there is no corresponding entity for w_i, the entity e_i is assigned a zero vector, as illustrated in Figure 3.

Subsequently, the token sequence $W = \{w_0, ..., w_n\}$ and the entity token sequence $E = \{e_0, ..., e_m\}$ are both processed through a Transformer Encoder[39], aligning them to the same dimension.

$$\begin{aligned} \{w'_0, ..., w'_n\} &= T\text{–}Encoder(\{w_0, ..., w_n\}), \\ \{e'_0, ..., e'_n\} &= T\text{–}Encoder(\{e_0, ..., e_n\}). \end{aligned} \tag{1}$$

Then, to inject knowledge $E = \{e'_0, ..., e'_n\}, e'_i \in R^{D'}$ into semantic sentence $W = \{w'_0, ..., w'_n\}, w'_i \in R^{D'}$, we use following equation:

$$\overline{w}_i = \sigma(W_w w'_i + W_e e'_i + b). \tag{2}$$

where, W_w and W_e represent learnable parameters, and b denotes the bias. In instances where a word token lacks a corresponding entity, the entity embedding is set as $e_i = [0, ..., 0]$, ensuring that Equation 2 equals:

$$\overline{w}_i = \sigma(W_w w'_i + b). \tag{3}$$

Following the acquisition of the integrated token embeddings $\overline{W} = \{\overline{w} = \overline{w}_0, ..., \overline{w}_n\}, \overline{w}_i \in R^{\overline{D}}$, we proceed with the feed-forward

[2]https://services.d4science.org/group/tagme/

process through the Transformer Encoder to attain the final output embedding.

$$\{\mathbf{w_0}, ..., \mathbf{w_n}\} = T\text{–}Encoder(\{\overline{w}_0, ..., \overline{w}_n\}) \quad (4)$$

The overarching framework for extracting textual features is depicted in Figure 3. We use a 12-layer BERT-base model (uncased) with a hidden size of 768 and 12 attention heads as our textual encoder. This provides a strong baseline for semantic modeling while enabling efficient integration with external knowledge embeddings.

3.2 Visual Features Extraction

To effectively capture the fine-grained attributes of the attached image, denoted as I, we adopt the Faster R-CNN approach as outlined in the bottom-up-attention model [2] and following the setup detailed. This process involves extracting the top-k ROIs (Regions of Interest) with the highest object detection scores. Subsequently, the embedding of each region object is obtained through average pooling, yielding f_i, which is then passed through a fully connected layer to generate the image representation denoted as V $=\{v_0, ..., v_k\}$, with $v_i \in \mathcal{R}^{D_V}$, as shown in Equation 5. The region list V is organized in descending order based on the detection scores. It is important to note that our model is trained on the Visual Genome dataset, which encompasses a diverse range of fine-grained classes, including "nose," "eye," "arms," "grass," "tree," and "beach".

$$v_i = W_f f_i + b_f, \quad (5)$$

Drawing inspiration from visual semantic reasoning [2], we construct a fully connected graph $G_V = \{V, R_V\}$, where V represents the set of embedding object regions, and R_V denotes the pairwise affinities between image regions in V, determined using Equation 6.

$$R_V(v_j, v_K) = (W_i v_i)^T (W_j v_j), \quad (6)$$

where W_i and W_j is learnable parameter. The relation score R signifies the semantic correlation between object regions within the image.

To perform semantic reasoning over detected image regions, we construct a visual graph where nodes represent object region embeddings $V = \{v_0, \ldots, v_k\}$, and edges encode semantic affinities between them. Specifically, we define an affinity matrix $R_V \in \mathbb{R}^{k \times k}$, where each element $R_V(i, j)$ quantifies the semantic similarity between region v_i and v_j, as computed in Equation 6.

We apply a Graph Convolution Network (GCN) [19] over this structure to propagate and refine region-level features. The output embedding for each region is computed by aggregating information from its neighbors, as shown in Equation 7:

$$V' = W_R(R_V V W_G) + V, \quad (7)$$

where $W_G \in \mathbb{R}^{D_V \times D'}$ and $W_R \in \mathbb{R}^{D' \times D_V}$ are learnable weight matrices. The matrix product $R_V V W_G$ performs neighborhood aggregation: each node receives a weighted sum of its neighbors' transformed features. This is followed by a linear transformation and a residual connection to preserve the original embedding. The resulting output $V' = \{v'_1, ..., v'_k\}, v'_i \in \mathbb{R}^{D_V}$ captures both local and relational context across object regions.

This GCN-based encoding enables the model to reason jointly over all image regions and capture their interactions, improving visual context understanding for multimodal alignment.

Subsequently, we proceed to feed forward V' through the Transformer Encoder, followed by utilizing average pooling to obtain the global image representation.

$$\mathbf{V} = AvgPool(T\text{–}Encoder(\{v'_0, ..., v'_n\})), \mathbf{V} \in R^{\mathbf{D}} \quad (8)$$

The overall framework for visual features extraction is illustrated in Figure.4.

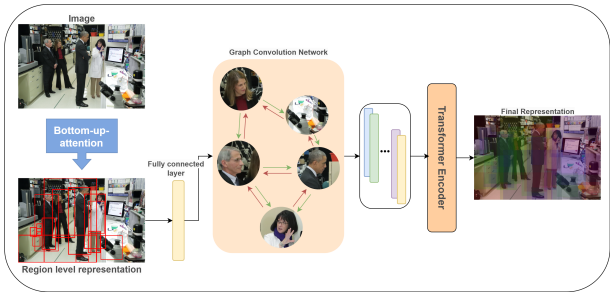


Figure 4: Overview of the Visual Features Extraction Framework. Initially, the image undergoes inference through the bottom-up-attention model [2] to acquire a list of region-level representations. Subsequently, the process continues with feedforward operations through a fully connected layer and a graph convolution network [19] to engage in region relationship reasoning. Following this, the list of enhanced region-level embeddings, facilitated by semantic relations, is feed-forwarded through a Transformer Encoder to perform global semantic reasoning and attain the final representation.

3.3 Combine visual and textual result to predict final output

3.3.1 Cheapfakes Detection. As we mention in Section 2, the training set of the cheapfakes dataset lacks fake/out-of-context cases. This characteristic prevents the application of the conventional approach of training and evaluating on the same dataset. In line with the setting of La et al. [20], we adopt a similar approach to facilitate comparisons. Specifically, we focus on two crucial aspect: the truthfulness of the information/caption and the correlation between the information/caption with the attached image. We train two seperate models for each of these aspect.

First, to assess the veracity of the information or caption, we focus on the final output $\mathbf{w_0}$ obtained from the set $\{\mathbf{w_0}, ..., \mathbf{w_n}\}$, which corresponds to the [CLS] token and serves as a representation of the information or caption. This representation vector is then employed for training a language model. Following this, the representation vector is fed forward through a fully connected layer, and a softmax activation function is applied for further processing. Formally,

$$\mathbb{P}_{NOOC} = Softmax(G_a(\mathbf{w_0}); \theta_a), \quad (9)$$

where, G_f denotes a linear layer with learnable parameters θ_f, and $\mathbb{P}_{NOOC}$ represents the probability score for "not-out-of-context" prediction derived from the trained language model. In our proposed language model, we employ a binary cross-entropy loss as

the objective function, as shown in Equation 10:

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N} y_i \cdot log(\mathbb{P}_{NOOC}) + (1 - y_i) \cdot log(1 - \mathbb{P}_{NOOC}), \tag{10}$$

Next, to evaluate the correlation between the image and the caption, we initiate the process with the caption by employing Named Entity Recognition (NER) for entity mapping to their corresponding categories. This step is essential to generalize the caption, as the knowledge graph contains an extensive quantity of entities. Subsequently, the preprocessed text is fed through a Gated Recurrent Unit (GRU) [9] to map the caption into the same dimensional vector space as $\mathbf{V}$ discussed in Section 3.2. The representation of the processed caption is denoted as $C \in R^{\mathbf{D}}$.

Subsequently, for our objective function, we employ a hinge-based triplet ranking loss, as depicted in Equation 11. This loss function is crucial in guiding our model to learn how to rank image-caption pairs effectively.

$$\mathcal{L}_M = [\alpha - S(\mathbf{V}, \mathrm{C}) + S(\mathbf{V}, \hat{\mathrm{C}})]_+ + [\alpha - S(\mathbf{V}, \mathrm{C}) + S(\hat{\mathbf{V}}, \mathrm{C})]_+, \tag{11}$$

where α is margin parameter, $[loss]_+ \equiv max(loss, 0)$, $S(\mathbf{V}, \mathrm{C})$ is computed by dot similarity function between $\mathbf{V}$ and C, $\hat{\mathrm{C}} = argmax_{C' \neq C} S(\mathbf{V}, \mathrm{C})$ and $\hat{\mathbf{V}} = argmax_{\mathbf{V}' \neq \mathbf{V}} S(\mathbf{V}', \mathrm{C})$ is hardest negative sample for a matching pair $(\mathbf{V}, \mathrm{C})$. For computational efficiency, the hardest negative sample is choose with in mini-batch.

Following the approach described in [20], after training the image-text matching model, we employ Equation 12 to normalize the correlation score between the caption and the image. This normalized score provides a measure of the correlation between the matching information/caption and the image in comparison to other non-matching samples.

$$\hat{\mathrm{S}}(\mathbf{V}, \mathrm{C}) = S(\mathbf{V}, \mathrm{C}) - \frac{1}{2N}\sum_{r}^{N}[S(\mathbf{V}, \mathrm{C}_r) + S(\mathbf{V}_r, \mathrm{C})], \tag{12}$$

where $\hat{\mathrm{S}}$ define normalize matching score, r indicate random sample satisfy $C_r \neq C$, $\mathbf{V}_r \neq \mathbf{V}$. Therefore, to determine truthfulness of information, we use simple rule as we illustrated in Equation 13:

$$OOC = \begin{cases} True, \text{if } \mathbb{P}_{NOOC} < t_c \ \& \ \hat{\mathrm{S}}(\mathbf{V}, \mathrm{C}) < t_i \\ False, otherwise \end{cases} ,. \tag{13}$$

In a simplified manner, we consider a pair of information/caption and an attached image as being "out-of-context" only when both the confidence score $\mathbb{P}_{NOOC}$ and the normalized matching score $\hat{\mathrm{S}}(\mathbf{V}, \mathrm{C})$ are smaller than predefined thresholds t_c and t_i.

3.3.2 Fake news detection. In the case of fake news detection. We follow approaches similar to [36, 43] that are trained and validated on the same type of dataset. We employ a similar information fusion method as described in Equation 3. This involves integrating the global image representation with the [CLS] token from the final output obtained in Section 3.1. Similar to the cheapfakes detection, we proceed to feed the represent token obtain from fusing image and textual information features through a fully connected layer followed by a softmax activation to predict whether the given news or information is fake or real. Formally,

$$\mathbb{P}_{fake} = Softmax(G_d(\sigma(W_{\mathbf{w_0}}\mathbf{w_0} + W_{\mathbf{V}}\mathbf{V} + b)); \theta_d), \tag{14}$$

where the term G_d represents a linear layer with learnable parameters denoted as θ_d for the training and validation method. The probability score $\mathbb{P}_{fake}$ is the prediction score for classifying news as fake from the trained language model. Just as in adaptation learning, we employ binary cross-entropy loss as the objective function for our proposed language model:

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N} y_i \cdot log(\mathbb{P}_{fake}) + (1 - y_i) \cdot log(1 - \mathbb{P}_{fake}), \tag{15}$$

Utilizing a straightforward approach for training and integrating visual with textual features, we can assess the effectiveness of our proposal in comparison to other methods.

4 Experimental Results

In this section, we present and analyze the experimental outcomes obtained by applying our methods and comparing them to other techniques within the same testing environment and datasets.

4.1 Experiments Setup

4.1.1 Dataset. For the cheapfakes detection, we employed the COSMOS dataset. This dataset is specifically designed to focus on untruthful textual information or captions and discrepancies between textual and visual content. The training dataset exclusively contains in-context captioned images, while the evaluation dataset includes both in-context and out-of-context captioned images. The evaluation dataset is divided into two parts: Part (I) includes two captions and one image for each captioned image, with each pair always featuring at least one caption with true information. Part (II) consists of one caption and one image for each captioned image.

For extended evaluation in fake news detection, we conducted experiments on the FakenewsNet benchmark, a widely recognized public benchmark dataset for fake news detection. This dataset was sourced from two fact-checking platforms, Gossipcop and Politifact. Both platforms provide comprehensive information, including visual, textual, and social context details, such as user profiles, comments, reactions, and spatial and temporal information. Given that the benchmark considers cases where attached images can be fake, unreal, or altered, we have adjusted the prediction step, as discussed in Section 3.3. This modification enables us to compare our method with others and assess its effectiveness across various types of benchmark datasets.

4.1.2 Implementation Details. In terms of representing external knowledge, we make use of TransE [6], which has been pre-trained on the Wikidata5m dataset. The Wikidata5m dataset contains a substantial number of 4,594,485 entities and 20,614,279 factual triplets. It is important to note that the knowledge graph remains unchanged during its application to the fake news and cheapfakes detection model.

During the cheapfakes detection phase, our language model is trained using the Fake and Real News dataset[3]. In addition, our image-caption matching model is trained on the cheapfakes training dataset. Both of these datasets are randomly divided, with 80% of the data used for training and the remaining 20% for validation. To determine the thresholds for out-of-context detection, we tuned t_c

[3]https://www.kaggle.com/datasets/clmentbisaillon/fake-and-real-news-dataset

Method	Accuracy	Precision	Recall	F1-score
Spotfake [36]	0.535	0.525	0.531	0.528
Pham et al. [26]	0.597	0.591	0.632	0.611
EANN-Multimodal [43]	0.630	0.603	0.612	0.619
Seo et al. [31]	0.719	0.650	0.693	0.717
SBERT-WK [42]	0.770	0.724	0.857	0.785
MUICG [21]	0.793	0.789	0.800	0.794
TeGA [23]	0.794	-	-	-
COSMOS [3]	0.819	**0.852**	0.772	0.810
Vu et al. [41]	**0.829**	0.816	0.816	0.821
Ours - text based w/o KG	0.803	0.807	0.796	0.802
Ours - text based	0.820	0.827	0.810	0.818
Ours - multimodal based	0.826	0.828	**0.831**	**0.829**

Table 1: Task 1: The results. The best results are highlighted in bold.

Method	Public				Private
	Accuracy	Precision	Recall	F1-score	Accuracy
LSTM [16]	0.610	0.612	0.600	0.606	-
BERT [11]	0.690	0.694	0.680	0.687	-
Seo et. al. [31]	0.557	0.531	0.579	0.637	0.615
TeGA [23]	-	-	-	-	0.630
TVE-UA [38]	0.730	0.735	0.720	0.727	-
VSRN-TEBA [20]	0.760	0.750	0.780	0.765	-
Ours - text based w/o KG	0.790	0.809	0.760	0.784	-
Ours - text based	0.830	0.851	0.800	0.825	-
Ours - multimodal based	**0.850**	**0.857**	**0.840**	**0.849**	**0.675**

Table 2: Task 2: The results for both public and private datasets. The best results are highlighted in bold.

and t_i using a small validation set (10% of validation data). A grid search was conducted to find values that maximized the F1-score, resulting in $t_c = 0.46$ and $t_i = 0.12$.

In the context of extend evaluation on fake news detection, we follow the same configuration as employed by Amila et al. [35]. This involves splitting the Gossipcop and Politifact datasets, with 75% allocated for training and 25% for testing. This entire process is repeated five times, and the resulting performances are averaged to obtain the final results. We use Adam optimizer with learning rate 2e-4 and weight decay 1e-5 to train both proposal for cheapfakes and fake news detection. We evaluate the models using four key metrics: Accuracy, Precision, Recall, and F1-score.

4.2 Performance Analysis

Quantitative Results for Cheapfakes Detection - COSMOS benchmark: The results of Task 1 and Task 2 are presented in Table 1 and Table 2, respectively. It is evident that our approach has achieved outstanding performance, establishing itself as the state-of-the-art solution for both Task 1 and Task 2 within the COSMOS dataset.

In Task 1, the dataset comprises pairs of two captions and one image each. We have compared our method against two baseline approaches: (I) considering both captions and the image, and (II) using one caption and one image to evaluate information accuracy. Methods like Spotfake [5] and EANN [43] combine textual and visual information through concatenation. While EANN performs admirably on Gossipcop and Politifact datasets, it lags behind on the more challenging COSMOS dataset. Other approaches, such as SBERT-WK [42] and [26], leverage pre-trained knowledge from large textual datasets, achieving impressive results with up to 77% accuracy. La et al. [21] use image captioning to describe image content and infer abstract relationships between captions for news classification. The COSMOS baseline [3] compares information between two captions, utilizing the image-caption pair to filter out cases where both captions do not mention the same object/topic in the image. Seo et. al. [31] and TeGA [23] utilize stable diffusion with various pretrained model to estimate the correlation between two captions and image. Vu et. al. [41]. take advantage of text-image model and prompting to predict the output. In contrast, our method focuses on a single caption and image but still achieves remarkable performance, surpassing other approaches. Therefore, even when relying solely on textual features, our approach consistently outperforms its counterparts.

In Task 2, each dataset pair consists of only one caption and one image. This task is notably more challenging than Task 1, as it necessitates the direct verification of information accuracy rather than assessing correlations between two sets of information. We also include two established language modeling frameworks, LSTM [16] and BERT [11], which are trained on the "Fake and Real News" dataset to evaluate their performance based solely on language vocabulary. Additionally, we compare our method with recent approaches such as VSRN-TEBA [38], TVE-UA [38], TeGA [23], and a multi-stage deep learning approach [31], which incorporate multimodal reasoning for fake news detection. Our results demonstrate that while these methods improve performance by integrating textual and visual features, they still struggle with fine-grained object recognition and contextual alignment. Specifically, TeGA improves text-guided representation learning but lacks robustness in handling real-world multimodal inconsistencies, leading to semantic misalignment errors. The multi-stage deep learning approach refines text-image and image-image alignment but remains sensitive to noise in visual elements, reducing its accuracy in challenging verification tasks. In contrast, our method, leveraging Knowledge Graphs and fine-grained object recognition, achieves higher accuracy and robustness, particularly in handling complex multimodal inconsistencies. As seen in the table, our approach excels, setting a new state-of-the-art in both public and private Task 2 of the COSMOS dataset. This indicates the stability and superior performance of our proposal, particularly in handling out-of-context manipulations and misinformation detection.

FakenewsNet benchmark: As shown in Table 3, our comparative analysis of various fake news detection approaches reveals distinct performance patterns within different domains: (T) content text-based methods, (S) social context-based methods, and (M) multimodal methods. Notably, approaches rooted in social context, which make use of textual information and user comments, tend to outperform purely text-based methods. However, they fall behind multimodal approaches that combine both textual and visual information.

Our proposed method demonstrates significant advancements in fake news detection by effectively leveraging both textual and visual information through multimodal integration. Unlike traditional text-based methods that rely solely on linguistic patterns, our approach enhances contextual understanding by incorporating

Method	Type			Politifact				Gossipcop			
	T	S	M	Acc	Prec	Rec	F1	Acc	Prec	Rec	F1
LIWC[25]	✓			0.488	0.680	0.565	0.432	0.662	0.550	0.516	0.472
text-CNN[18]	✓			0.608	0.621	0.623	0.608	0.733	0.698	0.703	0.701
HAN[44]	✓			0.632	0.672	0.651	0.648	0.716	0.703	0.709	0.706
LLAMA2[37]	✓			0.848	0.865	0.780	0.820	0.825	0.811	0.872	0.840
HPNF[32]		✓		0.697	0.692	0.683	0.687	0.721	0.703	0.689	0.695
CSI[30]		✓		0.734	0.774	0.571	0.658	0.753	0.748	0.764	0.756
AE[34]		✓		0.784	0.783	0.774	0.779	0.834	0.828	0.802	0.812
HPNF+LIWC[32]			✓	0.704	0.723	0.708	0.716	0.734	0.715	0.706	0.708
SAFE[46]			✓	0.793	0.782	0.771	0.775	0.831	0.822	0.798	0.806
EANN[43]			✓	0.804	0.808	0.794	0.798	0.836	0.812	0.815	0.813
MFAN[45]			✓	0.840	0.846	0.787	0.815	0.822	0.861	0.888	**0.875**
CAFE[8]			✓	0.848	0.857	**0.850**	0.853	0.832	0.804	**0.903**	0.851
Ours - text w/o KG	✓			0.782	0.812	0.776	0.793	0.803	0.703	0.726	0.714
Ours - text only	✓			0.806	0.830	0.806	0.818	0.826	0.751	0.732	0.741
Ours - multimodal			✓	**0.851**	**0.881**	0.841	**0.860**	**0.843**	**0.865**	0.868	0.867

Table 3: Fake news detection performance on Politifact and Gossipcop datasets. Baselines are grouped as: text (T), social context (S), and multimodal (M). Best results are in bold.

fine-grained visual features. This enables superior semantic reasoning, making our model more adept at identifying manipulated content.

Compared to large language models like LLAMA2 [37], which achieve 0.848 (Politifact) and 0.825 (Gossipcop), our text-based model performs competitively with 0.806 and 0.826, respectively highlighting the benefit of external knowledge via knowledge graphs for factual grounding.

Additionally, our method excels in achieving high precision and generalizability, ensuring that false positives are minimized while maintaining strong accuracy across diverse datasets. Compared to existing multimodal approaches, our model achieves strong accuracy and robustness, making it highly effective for real-world misinformation detection scenarios.

Method	Politifact	Gossipcop
ResNet-50 [15]	0.597	0.645
ViT-B-32 [12]	0.611	0.671
Our's - image based	0.605	0.659
Our's text based + ResNet-50	0.845	0.829
Our's text based + ViT-B-32	0.851	0.841
Our's - multimodal based	0.851	0.845

Table 4: Performance comparison of different methods across all datasets.

To better analyze the effectiveness of our method, we categorize results into text-only and multimodal-based settings. In the text-only setting, our model outperforms other baselines. Notably, removing entity embeddings led to an accuracy drop of up to 2.4%, underscoring the importance of external knowledge. Ablation studies (Table 4) further show that while ViT-B-32 performs best in isolation, our Bottom-Up Attention encoder is more effective when combined with text. This is due to its superior alignment of small objects and contextual elements with textual descriptions. Unlike ViT-B-32, which captures global features, Bottom-Up Attention enables fine-grained object-level reasoning, improving semantic alignment and enhancing cheapfake detection in multimodal settings.

5 Conclusions

This paper presents a robust framework integrating visual, syntactic, lexical, linguistic, and external knowledge for cheapfake detection. Unlike prior approaches, our method is evaluated on multiple datasets, achieving competitive results on FakeNewsNet and setting a new state-of-the-art on COSMOS.

By leveraging fine-grained visual features and external knowledge, the approach mitigates overfitting and enhances cross-domain reliability. Our results highlight the value of multimodal reasoning in improving fake news detection accuracy.

References

[1] Sahar Abdelnabi, Rakibul Hasan, and Mario Fritz. 2022. Open-domain, content-based, multi-modal fact-checking of out-of-context images via online resources. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 14940–14949.

[2] Peter Anderson, Xiaodong He, Chris Buehler, Damien Teney, Mark Johnson, Stephen Gould, and Lei Zhang. 2018. Bottom-up and top-down attention for image captioning and visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 6077–6086.

[3] Shivangi Aneja, Chris Bregler, and Matthias Niessner. 2023. COSMOS: Catching Out-of-Context Image Misuse Using Self-Supervised Learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 14084–14092.

[4] Kevin Aslett, Zeve Sanderson, William Godel, Nathaniel Persily, Jonathan Nagler, and Joshua A Tucker. 2024. Online searches to evaluate misinformation can increase its perceived veracity. *Nature* 625, 7995 (2024), 548–556.

[5] Pranav Bharadwaj and Zongru Shao. 2019. Fake news detection with semantic features and text mining. *International Journal on Natural Language Computing (IJNLC) Vol* 8 (2019).

[6] Antoine Bordes, Nicolas Usunier, Alberto Garcia-Duran, Jason Weston, and Oksana Yakhnenko. 2013. Translating embeddings for modeling multi-relational data. *Advances in neural information processing systems* 26 (2013).

[7] James Boyko, Joseph Cohen, Nathan Fox, Maria Han Veiga, Jennifer I Li, Jing Liu, Bernardo Modenesi, Andreas H Rauch, Kenneth N Reid, Soumi Tribedi, et al. 2023. An interdisciplinary outlook on large language models for scientific research. *arXiv preprint arXiv:2311.04929* (2023).

[8] Yixuan Chen, Dongsheng Li, Peng Zhang, Jie Sui, Qin Lv, Lu Tun, and Li Shang. 2022. Cross-modal ambiguity learning for multimodal fake news detection. In *Proceedings of the ACM web conference 2022*. 2897–2905.

[9] Junyoung Chung, Caglar Gulcehre, KyungHyun Cho, and Yoshua Bengio. 2014. Empirical evaluation of gated recurrent neural networks on sequence modeling. *arXiv preprint arXiv:1412.3555* (2014).

[10] Nadia K Conroy, Victoria L Rubin, and Yimin Chen. 2015. Automatic deception detection: Methods for finding fake news. *Proceedings of the association for information science and technology* 52, 1 (2015), 1–4.

[11] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* (2018).

[12] Alexey Dosovitskiy. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929* (2020).

[13] Lisa Fazio. 2020. Out-of-context photos are a powerful low-tech form of misinformation. *The Conversation* 14 (2020).

[14] Sunil Gundapu and Radhika Mamidi. 2021. Transformer based automatic COVID-19 fake news detection system. *arXiv preprint arXiv:2101.00180* (2021).

[15] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 770–778.

[16] Sepp Hochreiter and Jürgen Schmidhuber. 1997. Long short-term memory. *Neural computation* 9, 8 (1997), 1735–1780.

[17] Dhruv Khattar, Jaipal Singh Goud, Manish Gupta, and Vasudeva Varma. 2019. Mvae: Multimodal variational autoencoder for fake news detection. In *The world wide web conference*. 2915–2921.

[18] Yoon Kim. 2014. Convolutional neural networks for sentence classification. *arXiv preprint arXiv:1408.5882* (2014).

[19] Thomas N Kipf and Max Welling. 2016. Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907* (2016).

[20] Tuan-Vinh La, Minh-Son Dao, Quang-Tien Tran, Thanh-Phuc Tran, Anh-Duy Tran, and Duc-Tien Dang-Nguyen. 2022. A Combination of Visual-Semantic Reasoning and Text Entailment-based Boosting Algorithm for Cheapfake Detection. In *Proceedings of the 30th ACM International Conference on Multimedia*. 7140–7144.

[21] Tuan-Vinh La, Quang-Tien Tran, Thanh-Phuc Tran, Anh-Duy Tran, Duc-Tien Dang-Nguyen, and Minh-Son Dao. 2022. Multimodal Cheapfakes Detection by Utilizing Image Captioning for Global Context. In *Proceedings of the 3rd ACM Workshop on Intelligent Cross-Data Analysis and Retrieval*. 9–16.

[22] Nico Lang, Vésteinn Snæbjarnarson, Elijah Cole, Oisin Mac Aodha, Christian Igel, and Serge Belongie. 2024. From Coarse to Fine-Grained Open-Set Recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 17804–17814.

[23] Anh-Thu Le, Minh-Dat Nguyen, Minh-Son Dao, Anh-Duy Tran, and Duc-Tien Dang-Nguyen. 2024. TeGA: A Text-Guided Generative-based Approach in Cheapfake Detection. In *Proceedings of the 2024 International Conference on Multimedia Retrieval*. 1294–1299.

[24] Grace Luo, Trevor Darrell, and Anna Rohrbach. 2021. Newsclippings: Automatic generation of out-of-context multimodal media. *arXiv preprint arXiv:2104.05893* (2021).

[25] James W Pennebaker, Ryan L Boyd, Kayla Jordan, and Kate Blackburn. 2015. *The development and psychometric properties of LIWC2015*. Technical Report.

[26] Kha-Luan Pham, Manh-Thien Nguyen, Anh-Duy Tran, Minh-Son Dao, and Duc-Tien Dang-Nguyen. 2023. Detecting Cheapfakes using Self-Query Adaptive-Context Learning. In *Proceedings of the 4th ACM Workshop on Intelligent Cross-Data Analysis and Retrieval*. 60–63.

[27] Long-Khanh Pham, Hoa-Vien Vo-Hoang, and Anh-Duy Tran. 2024. A Generative Adaptive Context Learning Framework for Large Language Models in Cheapfake Detection. In *Proceedings of the 2024 International Conference on Multimedia Retrieval*. 1288–1293.

[28] Peng Qi, Zehong Yan, Wynne Hsu, and Mong Li Lee. 2024. SNIFFER: Multimodal Large Language Model for Explainable Out-of-Context Misinformation Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 13052–13062.

[29] Sijia Qian, Cuihua Shen, and Jingwen Zhang. 2023. Fighting cheapfakes: using a digital media literacy intervention to motivate reverse search of out-of-context visual misinformation. *Journal of Computer-Mediated Communication* 28, 1 (2023), zmac024.

[30] Natali Ruchansky, Sungyong Seo, and Yan Liu. 2017. Csi: A hybrid deep model for fake news detection. In *Proceedings of the 2017 ACM on Conference on Information and Knowledge Management*. 797–806.

[31] Jangwon Seo, Hyo-Seok Hwang, Jiyoung Lee, Minhyeok Lee, Wonsuk Kim, and Junhee Seok. 2024. A Multi-Stage Deep Learning Approach Incorporating Text-Image and Image-Image Comparisons for Cheapfake Detection. In *Proceedings of the 2024 International Conference on Multimedia Retrieval*. 1312–1316.

[32] Kai Shu, Deepak Mahudeswaran, Suhang Wang, and Huan Liu. 2020. Hierarchical propagation networks for fake news detection: Investigation and exploitation. In *Proceedings of the international AAAI conference on web and social media*, Vol. 14. 626–637.

[33] Kai Shu, Amy Sliva, Suhang Wang, Jiliang Tang, and Huan Liu. 2017. Fake news detection on social media: A data mining perspective. *ACM SIGKDD explorations newsletter* 19, 1 (2017), 22–36.

[34] Amila Silva, Yi Han, Ling Luo, Shanika Karunasekera, and Christopher Leckie. 2020. Embedding Partial Propagation Network for Fake News Early Detection.. In *CIKM (Workshops)*, Vol. 2699.

[35] Amila Silva, Ling Luo, Shanika Karunasekera, and Christopher Leckie. 2021. Embracing domain differences in fake news: Cross-domain fake news detection using multi-modal data. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 35. 557–565.

[36] Shivangi Singhal, Rajiv Ratn Shah, Tanmoy Chakraborty, Ponnurangam Kumaraguru, and Shin'ichi Satoh. 2019. Spotfake: A multi-modal framework for fake news detection. In *2019 IEEE fifth international conference on multimedia big data (BigMM)*. IEEE, 39–47.

[37] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288* (2023).

[38] Quang-Tien Tran, Thanh-Phuc Tran, Minh-Son Dao, Tuan-Vinh La, Anh-Duy Tran, and Duc Tien Dang Nguyen. 2022. A Textual-Visual-Entailment-based Unsupervised Algorithm for Cheapfake Detection. In *Proceedings of the 30th ACM International Conference on Multimedia*. 7145–7149.

[39] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems* 30 (2017).

[40] Hoa-Vien Vo-Hoang, Long-Khanh Pham, and Minh-Son Dao. 2024. Detecting Out-of-Context Media with LLaMa-Adapter V2 and RoBERTa: An Effective Method for Cheapfakes Detection. In *Proceedings of the 2024 International Conference on Multimedia Retrieval*. 1282–1287.

[41] Dang Vu, Minh-Nhat Nguyen, and Quoc-Trung Nguyen. 2024. Enhancing Cheapfake Detection: An Approach Using Prompt Engineering and Interleaved Text-Image Model. In *Proceedings of the 2024 International Conference on Multimedia Retrieval*. 1306–1311.

[42] Bin Wang and C-C Jay Kuo. 2020. Sbert-wk: A sentence embedding method by dissecting bert-based word models. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 28 (2020), 2146–2157.

[43] Yaqing Wang, Fenglong Ma, Zhiwei Jin, Ye Yuan, Guangxu Xun, Kishlay Jha, Lu Su, and Jing Gao. 2018. Eann: Event adversarial neural networks for multi-modal fake news detection. In *Proceedings of the 24th acm sigkdd international conference on knowledge discovery & data mining*. 849–857.

[44] Zichao Yang, Diyi Yang, Chris Dyer, Xiaodong He, Alex Smola, and Eduard Hovy. 2016. Hierarchical attention networks for document classification. In *Proceedings of the 2016 conference of the North American chapter of the association for computational linguistics: human language technologies*. 1480–1489.

[45] Jiaqi Zheng, Xi Zhang, Sanchuan Guo, Quan Wang, Wenyu Zang, and Yongdong Zhang. 2022. MFAN: Multi-modal Feature-enhanced Attention Networks for Rumor Detection.. In *IJCAI*, Vol. 2022. 2413–2419.

[46] Xinyi Zhou, Jindi Wu, and Reza Zafarani. 2020. : Similarity-Aware Multi-modal Fake News Detection. In *Pacific-Asia Conference on knowledge discovery and data mining*. Springer, 354–367.